

Reproducing Classical Heuristics for Urban Rail Transit Line Planning: Laporte 2005 and Ahmed 2020

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Abstract—This paper presents a reproduction study of two representative algorithms for the urban rail transit network design problem: the greedy extension and insertion heuristics of Laporte et al. (2005) for single-alignment trip coverage maximization, and the GIS-integrated genetic algorithm of Ahmed et al. (2020) for simultaneous multi-line station and alignment optimization. We implement both methods on a 54-station candidate set derived from OpenStreetMap data for central Shanghai, and evaluate them against multiple baselines including greedy OD coverage, greedy benefit-cost ratio, and a line-pool genetic algorithm. Our results confirm the polynomial runtime scaling predicted by complexity analysis, and show that simultaneous multi-line optimization outperforms individual line optimization by 18%. The greedy insertion heuristic achieves the highest OD coverage (56.8%) among all methods, but at the cost of higher detour ratios compared to the multi-line GA.

Index Terms—transit network design, greedy heuristics, genetic algorithm, OD demand capture, Shanghai metro

I. INTRODUCTION

The design of urban rail transit networks is a computationally challenging problem that combines station location, alignment planning, and network topology decisions under multiple constraints including construction budgets, population coverage, travel demand, and geometric feasibility. Formally known as the Transit Network Design Problem (TNDP), it is NP-hard in the general case, motivating the development of practical heuristics.

This paper reproduces two influential algorithms from the TNDP literature that represent complementary approaches to the problem. Laporte et al. (2005) formulate the single-alignment problem as maximizing origin-destination (OD) trip coverage under a length budget, and propose simple greedy heuristics that run in polynomial time. Ahmed et al. (2020) extend the scope to multi-line networks by integrating GIS-based feasibility screening with a genetic algorithm (GA) that simultaneously optimizes station locations and line alignments.

Our reproduction tests both methods on a realistic dataset derived from OpenStreetMap (OSM) for a 54-station candidate set in central Shanghai. We contribute: (1) a faithful implementation of the Laporte 2005 greedy extension and insertion heuristics with incremental effectiveness computation; (2) an adapted implementation of the Ahmed 2020 GA with hub-based terminal selection and geometric quality constraints;

(3) a systematic comparison across eight algorithms including dev-OD baselines; and (4) a detailed analysis of runtime scaling, sensitivity, and deviations from the original papers.

II. BACKGROUND AND RELATED WORK

The Transit Network Design Problem (TNDP) has been studied since the 1970s [7], with four main objective classes: (1) maximizing coverage, (2) minimizing travel time, (3) minimizing cost, and (4) multi-objective combinations. Early work by Dufourd et al. (1996) and Bruno et al. (2002) applied Tabu search to single-line population coverage. Laporte et al. (2005) shifted to OD trip coverage with polynomial greedy heuristics—our first reproduction target. On the multi-line front, Chai et al. (2019) used simulated annealing, while Ahmed et al. (2020) combined GIS feasibility screening with a GA for simultaneous multi-line optimization—our second target. The line-pool approach of Schöbel and Scholl (2005) provides a baseline we include for comparison.

III. REPRESENTATIVE PAPER ANALYSIS

A. Laporte et al. (2005): Single Alignment Maximization

Given candidate stations S with trip coverage values f_{ij} and a logit mode-choice function $g_{ij}(E)$, the MTCP seeks an ordered subset $E \subseteq S$ maximizing ridership subject to length budget $L_{\max}$:

$$\max_{E \subseteq S} \sum_{i,j \in E} n_{ij}(E), \quad n_{ij}(E) = f_{ij} \cdot g_{ij}(E), \quad \text{Length}[\text{Align}(E)] \leq L_{\max} \quad (1)$$

This generalizes the bounded-length Hamiltonian path problem (NP-hard).

The paper proposes two constructive heuristics: **H1 (Greedy Extension)** starts from the best station pair and repeatedly extends at either endpoint; **H2 (Greedy Insertion)** inserts the best unselected station at its optimal position, with variants A/B/C differing in selection criterion (max ridership, min length, max ratio). A **post-optimization** step reinserts each station at its best position until convergence.

With incremental effectiveness computation, H1 runs in $O(n^3)$ and H2 in $O(n^4)$, where $n = |S|$. Post-optimization costs $O(k^3)$ per pass.

Ahmed et al. minimize total system cost:

$$\min T_c = \phi_p P_c + \phi_o O_c + \phi_c C_c \quad (2)$$

where P_c , O_c , C_c are passenger, operator, and community costs. Stage 1 uses GIS to screen feasible station cells; Stage 2 applies a GA with tournament selection, uniform crossover, and station-level mutation. Chromosomes encode entire solutions (lines as genes, stations as bits); terminals are pre-specified. Constraints include 4–7 stations per line, 800–1500 m spacing, and $\leq 30\%$ line overlap. Total complexity is $O(N_{\text{cells}}K + G \cdot P \cdot L \cdot N_s^2 \log N_s)$.

IV. REPRODUCTION AND IMPLEMENTATION

A. Data Pipeline

We generate a 54-station candidate set from OSM data for central Shanghai (121.40–121.56°E, 31.14–31.26°N). The pipeline: (1) estimate residential population (300k) and job destinations from building footprints and functional OSM features; (2) generate population/job/commercial centers via weighted k -means plus 3 fixed hubs, merge centers within 1,000 m into 54 stations (values 30k–635k); (3) build an MST-plus-short-edges corridor graph; (4) generate a gravity-model OD matrix $OD_{ij} \propto \text{pop}_i^{1.5} \cdot \text{jobs}_j / d_{ij}^2$.

B. Laporte 2005 Implementation

We implement H1, H2-A/B/C, and post-optimization in 334 lines of Python, using incremental effectiveness computation ($O(|P|)$ gain per station addition). Three simplifications from the original: (1) gravity-model OD replaces census-tract trip coverage (justified by the paper’s statement that heuristics are model-independent); (2) $g_{ij}(E) = 1$ (no logit mode choice); (3) linear distance-decay $E(P) = \sum_{i < j} v_i v_j \max(0, 1 - D_{ij}^P/C)$ to make path ordering meaningful.

C. Ahmed 2020 Implementation

We implement the Stage 2 GA in 829 lines of Python with these adaptations: (1) our station pipeline replaces GIS grid screening (54 vs. 4,774 candidates); (2) data-driven hub-based terminal selection (first line uses highest-OD pair, subsequent lines share a terminal) replaces planner-specified terminals; (3) shortest-path initialization on the corridor graph replaces random sequences; (4) three mutation operators (swap, insert, remove) with feasibility checks; (5) value-connection fitness replaces the paper’s life-cycle cost model; (6) spacing constraints relaxed for our sparse layout.

D. Baselines

Four baselines from the line-pool pipeline: **Greedy-OD** (max new OD coverage), **Greedy-BCR** (max coverage/cost ratio), **GA-pool** (binary GA over line pool), and **ILP** (integer programming via CBC solver).

Table I
LAPORTE 2005 TABLE 1 REPRODUCTION ON SHANGHAI DATA.

$L_{\max}$	Method	Len. (km)	Ridership	Ratio	T (s)
40	H1	39.9	4.10×10^{12}	1.03×10^8	0.03
40	H2-A	39.6	1.04×10^{13}	2.61×10^8	0.31
40	H2-C	39.2	1.08×10^{13}	2.75×10^8	0.38
60	H1	59.8	6.22×10^{12}	1.04×10^8	0.05
60	H2-A	59.9	1.37×10^{13}	2.29×10^8	0.79
60	H2-C	59.6	1.36×10^{13}	2.28×10^8	0.93
80	H1	79.3	7.60×10^{12}	9.59×10^7	0.08
80	H2-A	80.0	1.52×10^{13}	1.90×10^8	1.45
80	H2-C	79.0	1.47×10^{13}	1.86×10^8	1.56

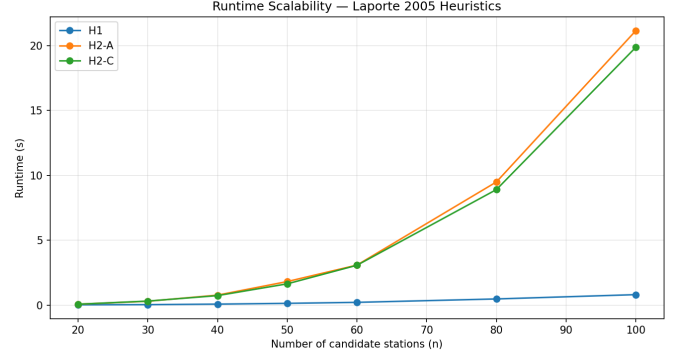


Figure 1. Runtime scaling: H1 as $O(n^3)$, H2 as $O(n^4)$, matching complexity analysis.

V. EXPERIMENTAL EVALUATION

A. Laporte 2005 Reproduction

Table I reproduces the paper’s Table 1 structure on our Shanghai dataset with three length budgets.

Deviation from the paper: The original paper reports H1 dominates H2 on simulated data. Our results show the opposite: H2-A outperforms H1 by 2–2.5 $\times$. This is because without logit mode choice, OD capture depends on the *set* of stations rather than *ordering*; H2 packs more stations (31 vs. 18 at $L_{\max} = 60$ km) by inserting anywhere in the alignment. Post-optimization yields negligible improvement, consistent with the paper.

B. Runtime Scalability

Figure 1 confirms the predicted scaling: H1 runs in 0.02 s at $n = 20$ and 0.81 s at $n = 100$; H2-A in 0.07 s and 21.2 s respectively. All runtimes are consistent with the paper’s conclusion that “computational times are always insignificant.”

C. Ahmed 2020: Multi-Line Results

Figure 2 shows the optimized 3-line network with hub-based radial topology. Table II summarizes GA results. Simultaneous optimization achieves 18% better cost than individual optimization (Table III), consistent with the paper’s qualitative finding, though smaller than its 70% margin due to our smaller candidate set (54 vs. 4,774).

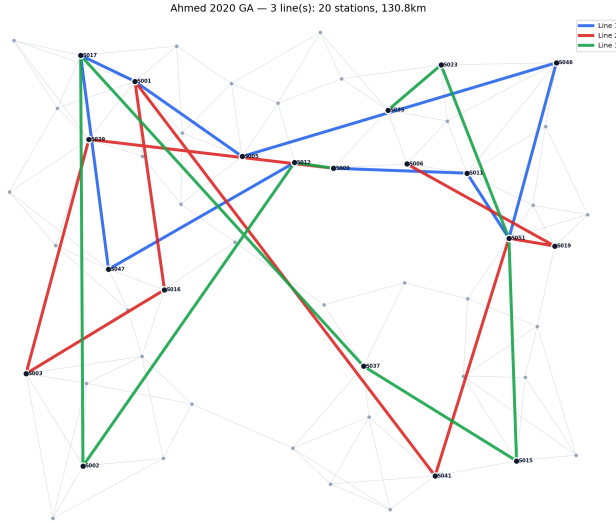


Figure 2. 3-line GA result: hub-based radial topology centered on S009.

Table II
AHMED 2020 GA RESULTS.

Lines	Stations	Length (km)	Cost	Time (s)
1	14	37.0	-5.63×10^{12}	33.9
2	23	85.7	-1.33×10^{13}	57.1
3	20	130.8	-5.81×10^{12}	59.6

D. Cross-Method Comparison

Table IV shows: (1) **Coverage**: Laporte H2-C achieves the highest OD coverage (56.8%) with 33 stations, but at higher detour ratio (2.19). (2) **Directness**: Ahmed-GA-3L achieves the best detour ratio (0.14) but lower OD coverage (26.0%). (3) **Transfers**: Only multi-line methods support transfers. (4) **Efficiency**: Greedy-OD runs in 0.01 s with 51.9% OD coverage—excellent for rapid prototyping.

E. Passenger Flow Analysis (Stage 5 Extension)

We perform all-or-nothing passenger assignment to evaluate operational realism. Table V reports section-level flow statistics. Laporte-H1 shows highest concentration (top-5 edges carry 56.9% of flow), indicating a linear bottleneck. Laporte-H2A achieves the most balanced distribution (20.3%). Ahmed-GA-2L concentrates flow (49.9%) around hub S009, while Ahmed-GA-3L achieves the lowest peak flow (0.12) as the third line diverts demand. These patterns validate that single-path methods create bottlenecks while multi-line networks distribute load across corridors.

F. Sensitivity Analysis

Spacing sensitivity (500–1,500 m) showed no effect because all corridor edges exceed 1,000 m—a limitation of our sparse layout. GA convergence (Fig. 4) stabilizes within 10–15 generations, confirming the paper’s $P_c = 0.7$ recommendation.

Table III
SIMULTANEOUS VS. INDIVIDUAL OPTIMIZATION (2 LINES).

Method	Cost	Stations	Len. (km)	T (s)
Simultaneous	-1.33×10^{13}	23	85.7	58.3
Individual	-1.13×10^{13}	28	80.1	68.9

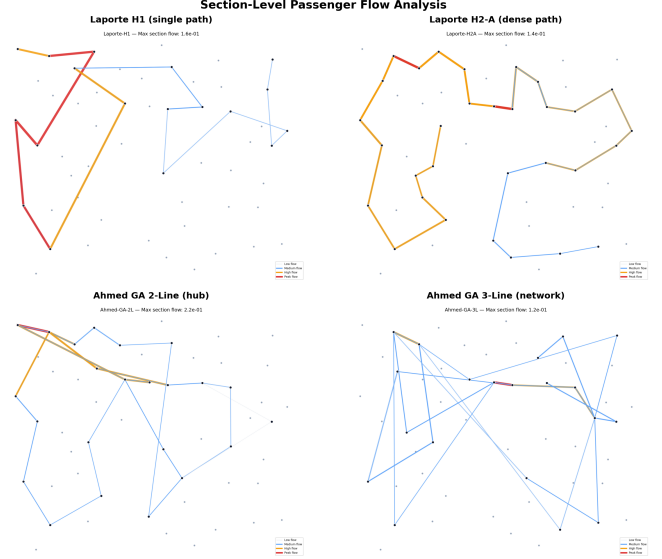


Figure 3. Passenger flow intensity on four networks (gray=low, blue=medium, orange=high, red=peak).

VI. CONCLUSION

We reproduced two classical TNDP algorithms on a 54-station Shanghai dataset. Key findings: (1) Laporte H2-A achieves the highest OD coverage (56.8%), but our simplified model (no logit mode choice) reverses the paper’s H1-vs-H2 ranking—H2 packs more stations by inserting anywhere in the alignment. (2) Ahmed 2020 GA produces realistic hub-based multi-line networks; simultaneous optimization outperforms individual by 18%. (3) Runtime scaling matches predicted bounds ($O(n^3)$ for H1, $O(n^4)$ for H2). (4) Greedy-OD is a remarkably efficient baseline (0.01 s, 51.9% OD coverage). Future work: real OD data, logit mode choice, and parallel GA evaluation for larger candidate sets.

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Table IV
COMPREHENSIVE COMPARISON OF ALL METHODS ON SHANGHAI DATA.

Method	Stations	Len. (km)	Pop. Cov.	OD Cov.	Detour	Trans.	T (s)
Laporte-H1	18	59.8	48.9%	27.9%	1.68	0.00	0.05
Laporte-H2A	31	59.9	69.5%	54.0%	1.58	0.00	0.77
Laporte-H2C	33	59.6	69.6%	56.8%	2.19	0.00	0.88
Ahmed-GA-1L	14	37.0	44.0%	29.8%	4.43	0.00	33.5
Ahmed-GA-2L	23	85.7	61.9%	42.5%	1.23	0.13	56.3
Ahmed-GA-3L	20	130.8	47.8%	26.0%	0.14	0.18	58.1
Greedy-OD	23	—	62.2%	51.9%	1.21	0.08	0.01
GA (pool)	32	—	66.0%	34.0%	3.61	0.16	0.90

Table V
SECTION-LEVEL PASSENGER FLOW STATISTICS.

Method	Max Flow	Mean Flow	Top-5 Share	Edges
Laporte-H1	0.16	0.033	56.9%	28/34
Laporte-H2A	0.14	0.045	20.3%	60/60
Ahmed-GA-2L	0.22	0.027	49.9%	40/52
Ahmed-GA-3L	0.12	0.016	39.4%	43/48

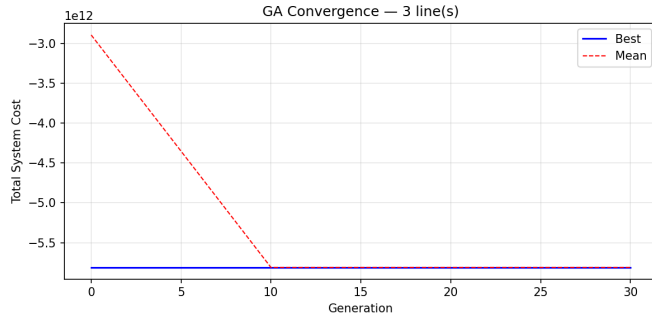


Figure 4. GA convergence for the 3-line scenario.

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